



Finance Case Research Club
Team Camelot



Individual Project: Osstem Implant

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I. Industry Overview.

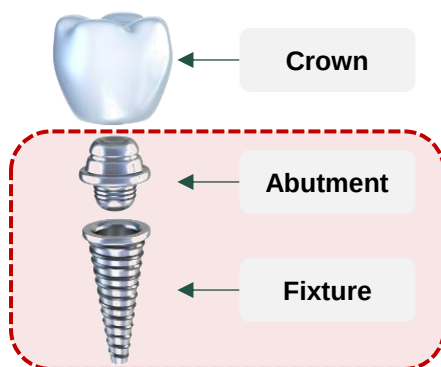
1. Dental Implant Industry Overview
2. Competitive Landscape

Dental Implant Industry Overview

The dental implant industry consists of upstream producers of implant abutments and fixtures with the rise of prominent domestic market trends regarding insurance regulation shifts and opportunities in China

Dental implant market overview

Dental implant components



Dental implant market

- Crown production is common among industry players; however, it is not a requirement for a dental implant producer
- The dental implant market consists of **players** who **manufacture** implant **abutments** and **fixtures**

Value chain

Upstream

Dental Implant Suppliers & Dental Equipment Manufacturers

Type

Major players

OSSTEM[®]
IMPLANT

Dentium

straumann

Midstream

Dental Laboratories

MODERN DENTAL

swiftdental group

Downstream

Dental Clinics & General/Private Hospitals

全民 Q&M

PacificSmiles DENTAL

Market trends: regulation changes for implant insurance

	Age	Num. of implants	Coverage
2014	75+ years old	Up to 2 implants	50% of cost covered
2018	65+ years old	Up to 2 implants	70% of cost covered
2024	65+ years old	Up to 4 implants ⁽¹⁾	70% of cost covered

Market trends: domestic firms penetrating Chinese market

Chinese implant market



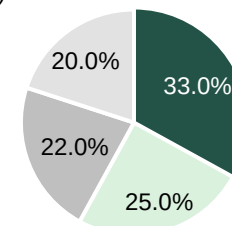
Blistering growth



The Chinese dental market expected to grow at **CAGR 12.3%** to over **US\$1.1bn** by 2030

Dental implant market share in China⁽²⁾

(%)



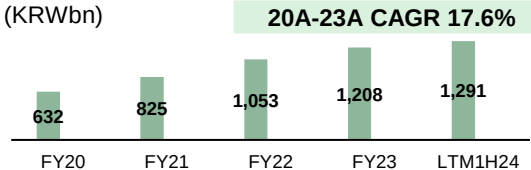
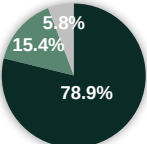
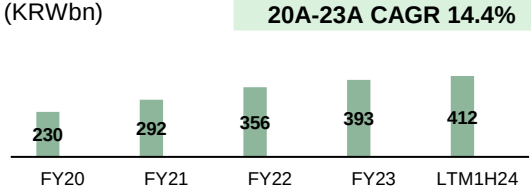
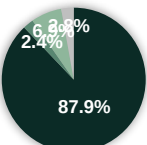
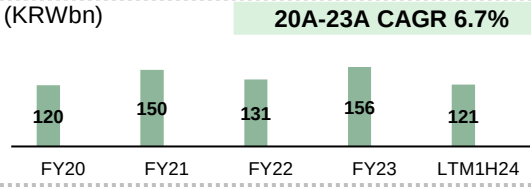
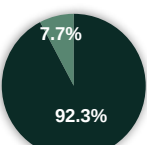
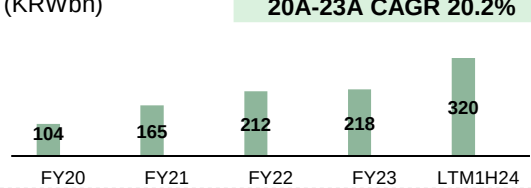
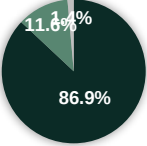
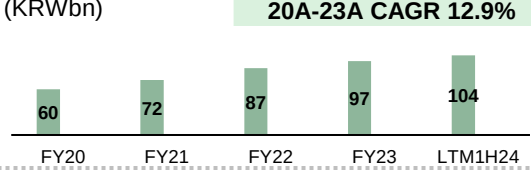
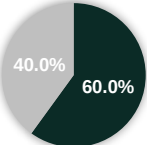
"More than 50% of the market is owned by domestic competitors"

Source: Company Filings, News Run

Notes: (1): Talks in progress to increase number of teeth to 4 as of 2024, (2) Dental implant market share in China as of 2022

Competitive Landscape

The domestic market is dominated by 5 firms – Osstem Implant, Dentium, Dio, MegaGen, and NeoBiotech – of which Osstem Implant records the highest revenue figures compared to its domestic peers

Company	Revenue	Product portfolio ⁽¹⁾	Notes
	(KRWbn) 20A-23A CAGR 17.6% 	(%)  - Dental Implants - Equipment - Others	- No. 1 position in terms of market share and revenue in the domestic market - Specializes in implant fixtures and surfaces
	(KRWbn) 20A-23A CAGR 14.4% 	(%)  - Dental Implants - Digital Dentistry - Dental Equipment - Others	Dentium is second in terms of revenue and focuses on cost-efficient implants catered to South Koreans' oral structures
	(KRWbn) 20A-23A CAGR 6.7% 	(%)  - Dental Implants - Digital Equipment	- Manufacturer of dental implants, toothpaste, and other dental equipment - Known for its DIO IMPLANT SYSTEM
	(KRWbn) 20A-23A CAGR 20.2% 	(%)  - Dental Implants - Digital Equipment - Others	A manufacturer of implants, with a noticeable presence and specialty in the dental prosthetics market
	(KRWbn) 20A-23A CAGR 12.9% 	(%)  - Dental Implants - Others	Has a smaller revenue profile than its peers, but operates in a niche market for implants utilizing snap technology

II. Case Study.

1. Company Snapshot
2. Business Overview
3. Deal Timeline

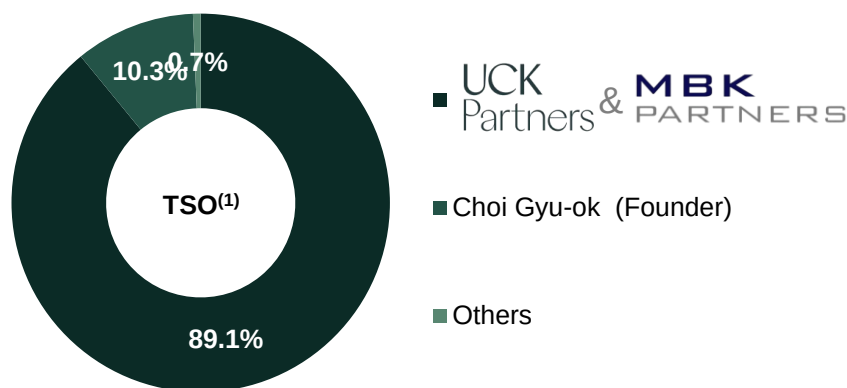
Company Snapshot

Osstem Implant is a global leader in the dental implant industry with a variety of dental products; in 2023, it was acquired by Dentistry Investment and was subsequently delisted from the KOSDAQ exchange

Company profile

- Established in 1997, Osstem Implant is a global total care service provider in the dental industry primarily focusing on dental implant production
- Osstem Implant went public in 2007 on the KOSDAQ exchange; however, it was delisted on August 2023 after the acquisition by Dentistry Investment, a consortium formed by MBK Partners and UCK Partners
- Osstem Implant is a global leader in the dental implant industry with the highest market share in the domestic dental implant market and 3rd highest in terms of market share in the global dental implant market
- Osstem Implant specializes in implant fixture surface technology, boasting their SOI (Super Osseointegration) fixture surface, the most effective product on the market in terms of recovery time and treatment period
- The company is also committed to global expansion, recently completing an acquisition of Brazilian dental implant supplier Implacil de Bortoli

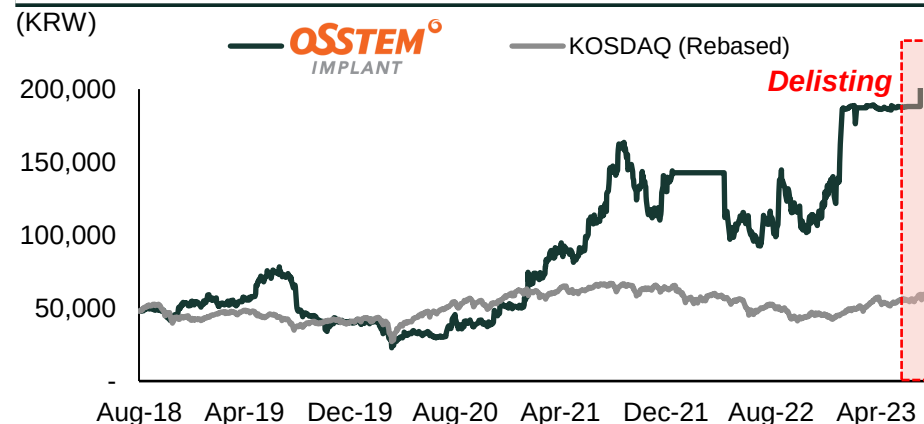
Shareholding structure



Key financials

(KRWbn)	FY20	FY21	FY22	FY23	LTM1H24
Revenue	632	825	1,053	1,208	1,291
YoY Growth(%)	-	30.6%	27.8%	14.7%	6.8%
Gross Profit	378	496	670	745	779
Margin (%)	59.9%	60.2%	63.6%	61.6%	60.3%
EBITDA	125	174	271	281	261
Margin (%)	19.8%	21.2%	25.7%	23.3%	20.2%
EBIT	98	143	235	243	219
Margin (%)	15.5%	17.4%	22.3%	20.1%	17.0%
Net Income	104	23	144	160	104
Margin (%)	16.4%	2.8%	13.7%	13.2%	8.0%
Total Assets	959	1,152	1,372	1,687	1,942
Total Liabilities	770	935	1,010	1,145	1,355
Net Debt	65	159	106	3	210

5Y stock price performance until delisting⁽²⁾



Source: Company Filings, KRX, News Run

Note: (1) Total Shares Outstanding, As of Jun 30, 2024;

(2) KOSPI rebased to Osstem Implant's stock price as of Aug 13, 2018; Delisted in Aug 14, 2023

Business Overview

Osstem Implant currently has three major businesses segments—dental implants manufacturing, dental equipment supplying, and dental services & IT software development—all of which Osstem Implant has been thriving in

Revenue breakdown⁽¹⁾

(KRWbn)

Dental implants
(78.9%)

524

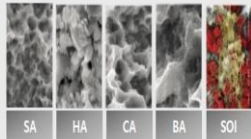
Dental equipment
(15.4%)

103

Others
(5.85%)

38

Product overview

	Key products	Key customers	Description
Dental implants	 TS, SS, US, MS fixtures⁽²⁾  Fixture surface	 Dental clinics  General hospitals  Laboratories	- Osstem Implant's unique designs are tailored to diverse oral and facial structures - Fixture surface options ensure optimal osseointegration with SOI (Super Osseointegration Implant) having up to 35% shorter healing time
Dental equipment	 TRIOS Oral Scanner  SM5 Implant Engine	 Dental clinics  General hospitals  Laboratories	- The TRIOS Oral Scanner is a state of the art wireless device that accurately scans the oral structure of patients in 3D - SM5 Implant Engine is a user-friendly device that helps streamline the implant surgery process
Dental services & IT	 OIC Education⁽³⁾  OneClick IT Program	 Dentists  Dental clinics	- OIC is a clinical implant education center training over 100,000 dentists globally - Osstem Implant is a pioneer in integrating IT software, being the <i>only domestic company</i> to provide implant & insurance software along with physical dental implants

Source: DART, News Run

Note: (1) Revenue breakdown as of 1H24, (2) TS: Transcendent Solution, SS: Success Solution, US: Universal Solution, MS: Micro Solution

(3) OIC: Osstem Implant training Center

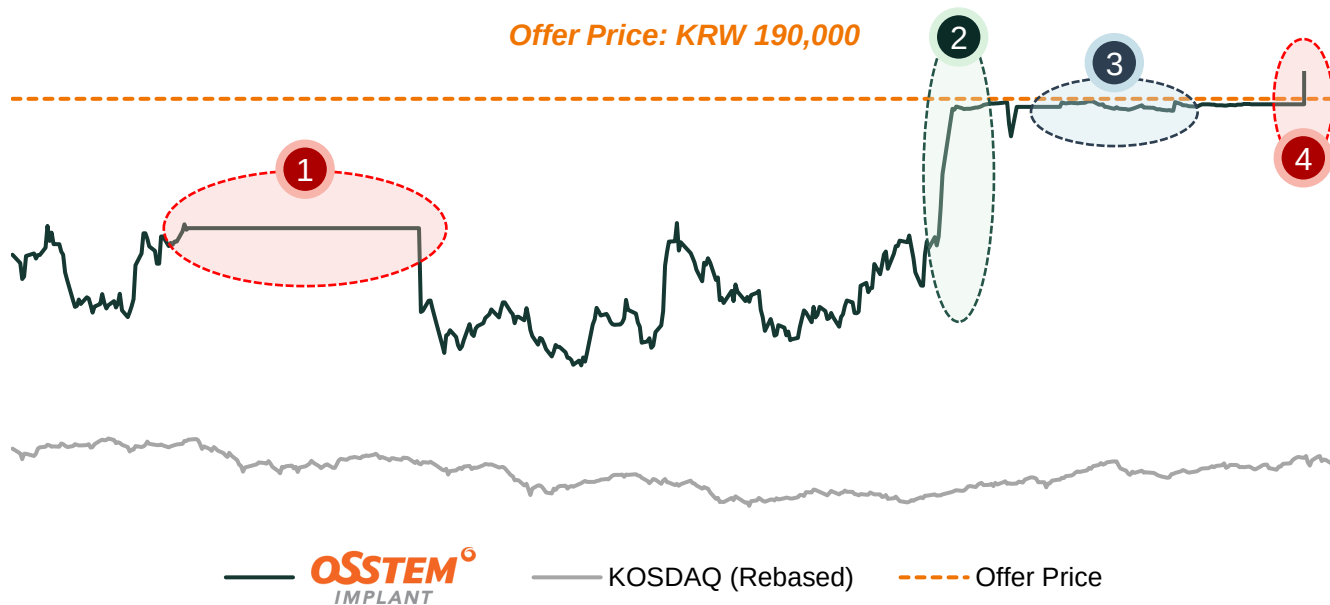
Deal Timeline

Dentistry Investment, a consortium between MBK Partners and Unison Capital Korea (UCK) Partners, acquired a majority share in Osstem Implant in 2023 and proceeded to delist the company from KOSDAQ

Osstem Implant stock price and deal timeline

(KRW)

Offer Price: KRW 190,000



Oct-21 Jan-22

Apr-22 Jul-22

Oct-22 Jan-23

Apr-23 Jul-23

1 Jan. 7, 2022

Embezzlement Case

- Employee embezzlement case arises, all trading on the KOSDAQ stops from Jan. 7 – Apr. 29, 2022

2 Feb. 24, 2023

Initial Acquisition

- The aftermath of the embezzlement case leads to MBK Partners & UCK's consortium to acquire a 65.1% stake for KRW 1.8tn via tender offer

3 Apr. 11, 2023

Secondary Acquisition

- 2nd acquisition was made by MBK & UCK to acquire a 6.7% stake for KRW 200bn via tender offer

4 Aug. 14, 2023

Delisting

- Osstem Implant delisted from the KOSDAQ
- Post-secondary acquisition, the SPC controlled 96.1% of shares⁽¹⁾ more than the consensus requirement⁽²⁾

Tender Offer Details

1st Tender offer

Tender offer: 71.8% stake for KRW 2.1tn for Jan. 25, 2023
Offer price: KRW 190,000 per share
Actual deal: 65.1% for KRW 1.8tn
Premium: (one week): 36.0%

2nd Tender offer

Tender offer: 10.6% stake for KRW 320bn on Mar. 22, 2023
Offer Price: KRW 190,000
Actual deal: 6.7% for KRW 300bn
Premium: (one week): 1.6%

Ownership structure

MBK PARTNERS

UCK Partners

Dentistry Investment (SPC)

stake: 83.6%⁽³⁾

OSSTEM[®] IMPLANT

Source: DART, Capital IQ, KRX, News Run

Notes: (1) 96.1% stake includes former majority shareholder Choi Gyu-ok's shares and treasury shares, (2) requirement for KOSPI is 95%

(3) Ownership as of June 2024: includes 71.81% from tender offers, 9.26% acquired from Choi Gyu-ok, and 2.53% via delisting trading

III. Investment Highlights

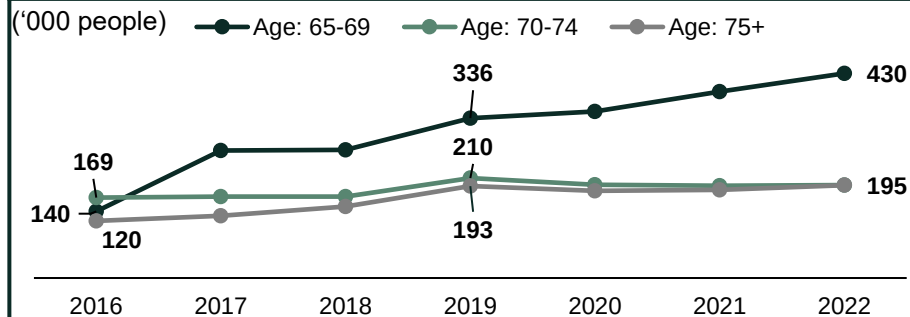
1. Socio-Economic Tailwinds for Strong Industry Growth
2. Dominant Market Share Propelled by Innovative Products
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Socio-Economic Tailwinds for Strong Industry Growth

The domestic dental implant industry is on a growth trajectory, coupled with the aging demographic and increased demand for implants by the elderly population, allowing the existing players to benefit

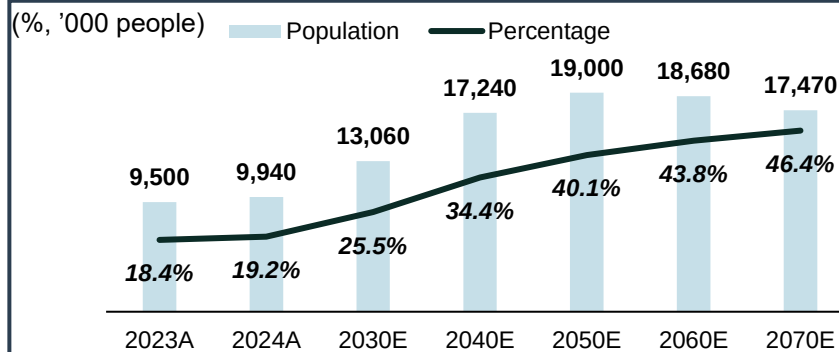
Demand drivers for growth of the domestic market

Number of dental implant patients by age



Demand for dental implants for the elderly is on an upward trajectory
with more inclusive and affordable insurance coverage

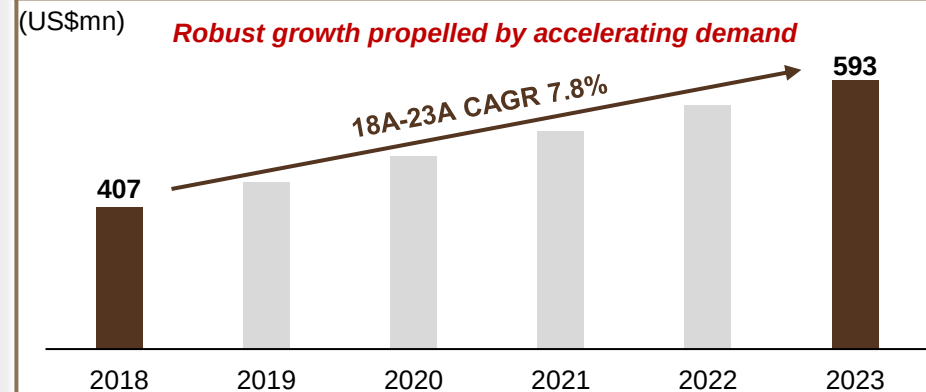
Aging population in South Korea⁽¹⁾



The aging population is a key revenue driver as the **primary age demographic of implant users tend to be elderly over 65 years old**

Domestic industry growth and key component

Dental implant growth in domestic market



Key component: high barriers to entry



Cost of replication

- The cost of replicating the R&D and technology needed to produce dental implants is high
- Established brand recognition among customers both domestically and globally



Oligopolistic market

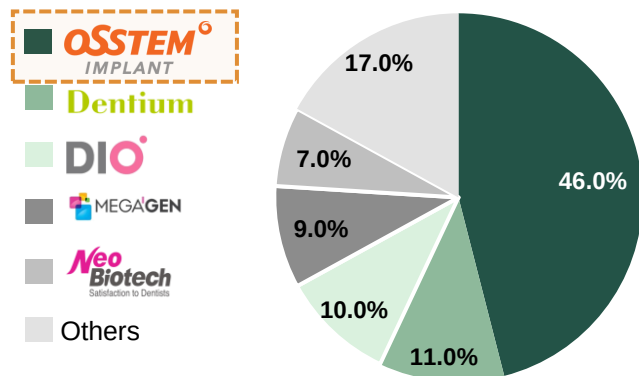
- 5 competitors dominate the majority of the market with there only being 10 active firms in the dental implant production market
- Medical related gov't regulations are strict

Dominant Market Share Propelled by Innovative Products

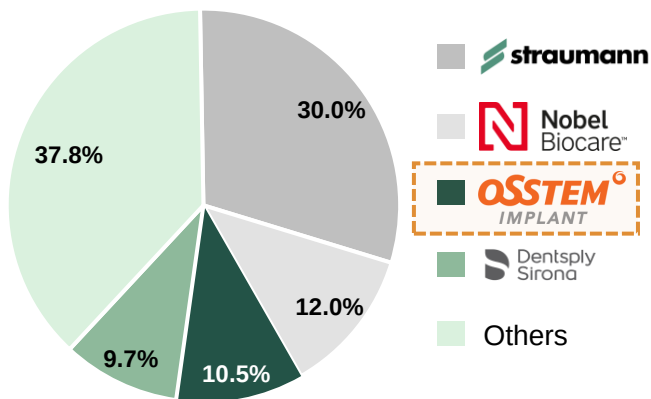
Osstem Implant has dominated the market both domestically and globally; its market dominance can primarily be attributed to its differentiated and innovative products with highly satisfactory reviews

Market leading position driven by differentiated products

Domestic dental implant market share



Global dental implant market share



Market share driver: product differentiation

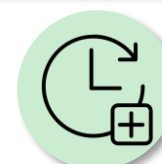
	OSSTEM IMPLANT	Dentium	DIO	MEGA GEN	Neo Biotech
1 Primary products	KSIII SA/SOI	Implantium	DIO Implant System	Anyridge	YK Link System
2 Main fixtures	5 Types	4 Types	3 Types	5 Types	3 Types
3 Surface technology	SOI	SLA	VUV (Vacuum UV)	XPEED	SLA
4 Treatment period	4-6 Weeks	8-12 Weeks	12-24 Weeks	12-16 Weeks	12-24 Weeks
5 Dentists' preference ⁽¹⁾	46.3%	18.2%	< 5.0%	8.9%	7.3%

Osstem Implant product highlights



Advanced technology

- Industry leading products such as SOI



Superior speed

- Quickest treatment period and 20-35% shorter healing time



Stellar quality

- Long lasting implants verified by industry professionals

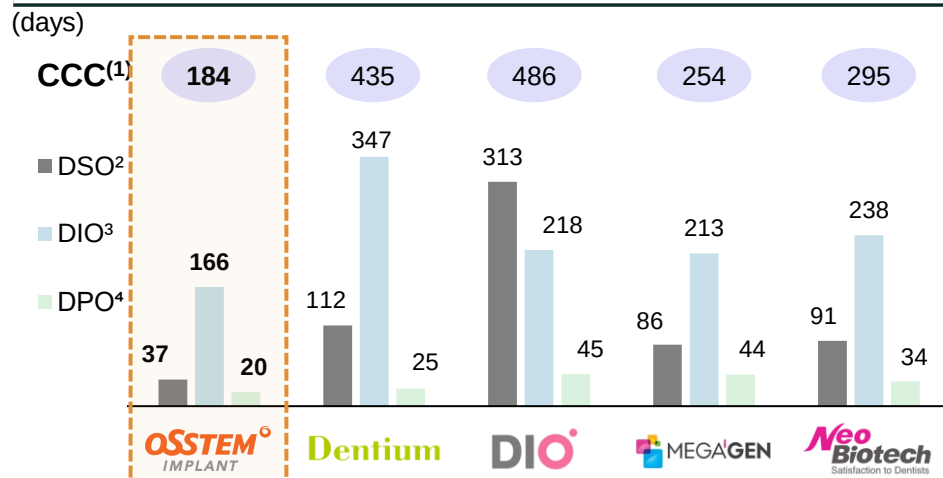
Source: DART, News Run, Market Intelligence

Notes: (1) Survey of domestic dentists' most preferred implant brand, conducted in 2024

Industry-leading Cash Generation and Financial Stability to Fund Growth

Osstem Implant has leveraged its market share to produce exceptional financials compared to its peers with substantially lower CCC and lower net debt to EBITDA than peers, signaling long-term stability and flexibility

Superior short-term cash generation capabilities



Days Sales Outstanding

- Low DSO compared to peer average: 150.3 days
- Downward influence over buyers is high; able to receive cash payments much sooner

Days Inventory Outstanding

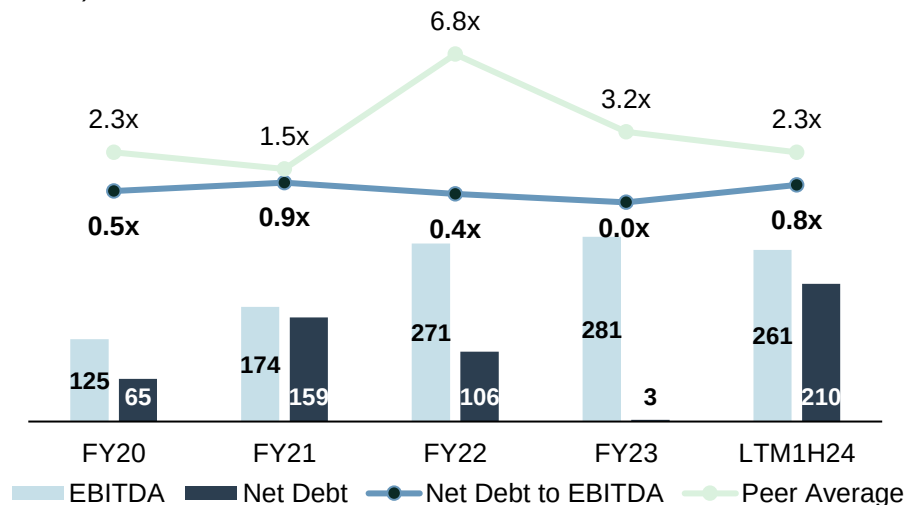
- Low DIO compared to peer average: 253.9 days
- Superior control over inventory stock

Cash Conversion Cycle

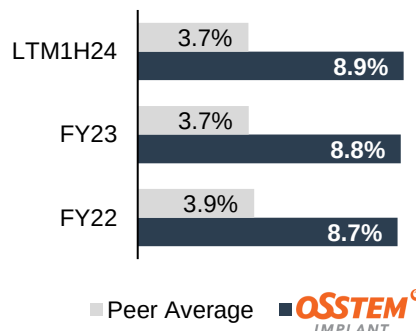
- Cash generation ability in the short term is greater than peers, translating to more cash on hand
- Recorded strong profitability: EBITDA Margin 20.2% for LTM 1H24

Financial stability & flexibility: low net debt-to-EBTIDA

(KRWbn)



R&D Expenditure to Revenue



- High EBITDA and cash
- Low levels of debt financing



- Highest level of R&D expenditure-to-revenue within industry

Source: DART, Capital IQ

Notes: (1) CCC: Cash Conversion Cycle, (2) DSO: Days Sales Outstanding, (3) DIO: Days Inventory Outstanding

(4) DPO: Days Payable Outstanding



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